

- 1 Fig. 1.1 shows a pBR322 vector which is commonly used to clone the human Growth Hormone (hGH) gene. In the formation of recombinant plasmid, it is known that restriction endonuclease *Bam*HI is usually used to digest the vector.

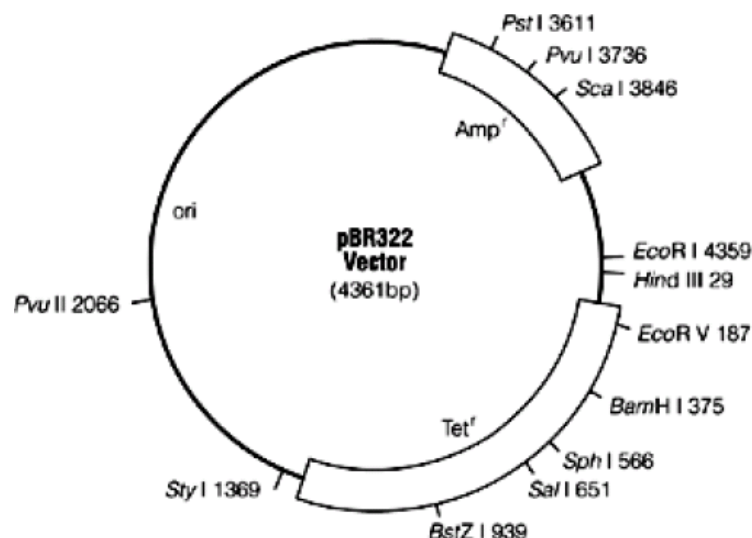


Fig. 1.1

- (a) Why is *Bam*HI suitable for the digestion of pBR322 vector? [1]
- only cleaves at one location / genetic marker;
 - allow for selection;
- (b) Outline the process to produce plasmids containing the hGH gene via genetic engineering. [4]

(a) Source of gene: [max 1]

Gene for hGH from genome

- Digest genomic DNA from cells of anterior pituitary gland;
- Use a suitable restriction enzyme / endonuclease;
- To isolate DNA fragment of hGH gene with complementary sticky ends;

OR

mRNA for hGH

- mRNA of hGH gene is extracted from cells of anterior pituitary gland;
- Reverse transcriptase is used to make a copy of complementary DNA (cDNA);
- Reverse transcriptase / DNA polymerase is used to synthesise the second DNA strand;
- Mentioning the use of linkers DNA (to create sticky ends since cDNA will be blunt);

(b) Creation of Recombinant plasmid: [max 3]

- plasmid/vector and cDNA / DNA are digested/cut/restricted with the same restriction enzyme/*Bam*HI;
- at (specific) restriction site/nucleotide sequence;
- same sticky/staggered ends created;
- sticky/staggered ends anneal via complementary base pairing;
- H-bonds formed between sticky ends of cDNA/DNA and plasmid/vector;
- DNA ligase is added to seal the nicks/ join the sugar-phosphate backbone;
- Formation of the phosphodiester bonds;

[Max 4]

(c) Following the formation of the plasmids containing the hGH gene, these recombinant plasmids are inserted into bacteria host cells. Name and describe the process involved. [3]

- **transformation;;**
- **Definition: A process where a host cell assimilates external DNA, resulting in genotypic and phenotypic changes / The genetic alteration of a host cell by artificial introduction of DNA;;**
- **Add calcium chloride to make bacteria competent;**
- **to increase permeability / create pores within cell membrane;**
- **subjected to heat shock / electroporation;**

[Max 3]

(d) Suggest how transformed bacteria with recombinant plasmid can be identified. [2]

- **Spread the transformed bacteria mixture on ampicillin plate;**
- **Bacterial cells (repeatedly divide and) grow to give rise to colonies;**
- **carry out replica-plating;**
- **plate the bacterial colonies on the ampicillin plate onto the tetracycline plate;**
- **compare the two plates;**
- **transformed bacteria → bacteria colonies present on the ampicillin plate but absent on the tetracycline plate;**

[Max 2]

In an investigation carried out by L, the detective, DNA was extracted from the remains of a victim whose body was blown beyond recognition. Restriction fragment length polymorphism (RFLP) analysis was carried out to determine his identity.

DNA samples of three pairs of couples who claimed to be the parents of the victim were isolated and analysed. A southern blot was prepared and probes, specific for certain RFLP markers in human beings, were added. Fig. 2.1 shows the banding patterns produced on the autoradiogram for analysis.

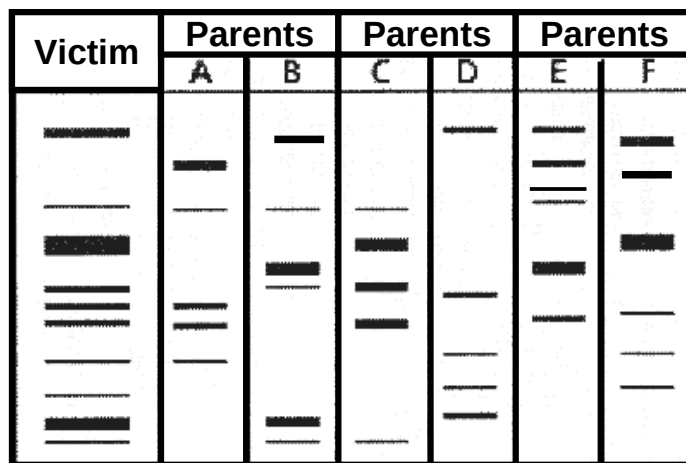


Fig. 2.1

(e) Explain the basis behind restriction fragment length polymorphism (RFLP). [2]

- Two organisms / individuals have different VNTR;;
- Differ in distance between restriction sites of a particular RE / number of restriction sites;;
- The length / number of the fragments produced will differ when the two DNA are digested with the same RE;;

[Max 2]

(f) Outline how genetic fingerprinting is carried out. [3]

- DNA is extracted from sample and cut by restriction enzyme (RE);
- Fragments separated by gel electrophoresis;
- according to their molecular size;
- Carry out (southern) blotting / transferred to a nylon/nitrocellulose membrane;
- Hybridize / Incubate with Single-stranded radioactive probes;
- which hybridise / anneal to complementary (target) DNA sequence / VNTR;
- Radioactive probes are detected using autoradiography / use of X-ray film / photographic film;

[Max 3]

(g) According to Fig. 2.1, which pair of parents is likely to be the parents of the victim? Explain your choice. [1]

- Parents C and D;
- Offspring inherit / gotten one set of chromosome for each parent;

[Total: 16]

- 2 Cystic fibrosis is an inherited genetic disorder that affects more than 8000 children and young adults in the UK. It affects individuals who carry two faulty copies of a gene called cystic fibrosis transmembrane regulator (CFTR), and results in mucus throughout the body becoming thicker and stickier than usual. This prevents several organs, in particular the lungs and digestive system, from functioning properly, and makes sufferers very susceptible to infections which can often be incurable or life-threatening. Currently, therapies used to treat CF only ease symptoms of the condition and those affected will typically die from lung or organ failure before the age of 40. 5

A recent study carried out at the University of California, Berkeley and the University of Iowa, US reported significant development in the field of gene therapy. The research documents the successful treating of cystic fibrosis (CF) in human lung tissue by using a highly infectious virus to deliver healthier genes into the affected cells. 10

The scientists modified a harmless virus, known as adeno-associated virus (AAV), into a highly infectious form several hundred times stronger than normal AAV. The new virus is more effective at entering lung cells than adenoviruses and better at evading the body's natural defence systems. 15

In clinical trials conducted, it has been shown that CF gene therapy is safe and to some extent works in patients; however, the duration of correction was found to be short-lived. Importantly, these early clinical trials demonstrated the efficacy of CF gene therapy and since then steady progress has been made towards gene therapy becoming a realistic therapeutic option for the disease. 20

(a) Explain what is meant by *gene therapy*. [2]

- **transfer/addition of one/more normal allele / gene into an individual's body cells/somatic cells;**
- **to correct a genetic defect (or boost resistance to a disease);**
- **via the use of vectors (viral and non-viral);**
- **for the insertion of a normal gene/allele / to restore/expressed normal phenotype;**

(b) Explain briefly how a mutation in the CFTR gene leads to cystic fibrosis. [2]

- **mutation on chromosome 7;**
- **nucleotides – CTT are removed;**
- **phenylalanine no longer coded for;**
- **non-functional CFTR protein (CFTR protein deficiency);**
- **CFTR protein now unable to transport chloride ions across membrane, leading to mucus buildup (AVP about CF);**

[Max 2]

(c) Suggest why adenoviruses were commonly used as viral vector for gene therapy to introduce normal CF gene (*line 17*). [1]

- **adenovirus causes colds and is very infectious of lungs / normally attacks respiratory epithelial cells, and injects its genome into the epithelial cells;;**

- (d) Explain how the adenovirus delivers the gene into lung cells. [2]
- **binding of viral glycoprotein (spike)/coat protein to specific receptor on plasma membrane of cells;**
 - **viruses is engulfed by receptor-mediated endocytosis;**
 - **plasma membrane invaginates and placing the virus in an endocytic vesicle / pinches off to form an endosome/intracellular vesicle;**
 - **Viral envelope fuses with endocytic vesicle membrane;**
 - **adenovirus / nucleocapsid is released into cytoplasm;**
 - **viral DNA enters nucleus (through pores in the nuclear envelope);**
- [Max 2]
- (e) Suggest why the modified adeno-associated virus is more effective than adenoviruses at entering lung cells. [1]
- **bind to different receptors;;**
 - **bind to a more plentiful receptor on the cell surface;;**
 - **an enhanced ability to make it past the cell surface membrane, slip past the lipid bilayer and reach the inside the cell;;**
- [Max 1]
- (f) It has been shown that CF gene therapy is safe and to some extent works in patients; however, the duration of correction was found to be short-lived resulting in patients having to repeat the treatment periodically (*line 19*). Explain why. [2]
- **Functional CFTR gene not integrated in genome / remains extrachromosomal;;**
 - **Short-lived nature / not stable and degraded after some time;;**
 - **cells have short lifespan & die/ immune system kills virus vector;;**
- [Max 2]
- (g) Explain why a person cured by gene therapy may still have children who suffer from the abnormality. [2]
- **gene therapy treats only target / somatic cells;;**
 - **gametes/germ cells are not affected/do not take up the healthy gene;;**

(h) Some people are concerned about using gene therapy to treat genetic disorders in humans. Give **two** arguments against the treatment of disorders by gene therapy. [2]

- **high cost plus reason e.g. compared with conventional treatments;;**
- **risk of stimulating the immune system (by the vector);;**
- **use of animals in (preliminary) testing;;**
- **other genes introduced which may have damaging effects/damage genome/damage other genes;;**
- **may induce a tumor if integrated in a tumor suppressor gene because insertional mutagenesis → cancer may result;;**
- **short-lived nature of gene therapy so patients will have to undergo multiple rounds of gene therapy;;**
- **immune system's enhanced response to invaders it has seen before makes it difficult for gene therapy to be repeated in patients;;**
- **possibility of viral vector recovering its ability to cause disease once in host;;**
- **inadequate understanding of biological interactions of vector and host;;**
- **conditions or disorders that arise from mutations in a single gene are the best candidates for gene therapy, but some the most commonly occurring disorders are caused by the combined effects of variations in many genes. Multigene disorders especially difficult to treat effectively using gene therapy;;**
- **changes to genetic make-up of individual/species/genome/future generations/germ line may affect normal development of offspring;;**

Reject - not ethical (without explanation)
(; = 1 mark each)

[Max 2]

[Total:14]

- 3 Kochia (*Kochia scoparia*) is a common weed in the Great Plains of the United States and Canada. Kochia affects crops by reducing the yield of wheat and corn. Kochia is difficult to control because of its leaf characteristics. The leaves either prevent herbicide contact or the herbicide spray droplets roll off the leaf.

Chemicals known as surfactants are detergents which help to aid the retention of herbicides on the leaves of the weeds. Research was undertaken to study the influence of surfactants on the effectiveness of three herbicides in aiding the control of Kochia.

Table 3.1 below shows the percentage biomass reduction of Kochia using three herbicides with different surfactants.

Table 3.1

Surfactant	Percentage Biomass Reduction		
	Bromoxynil	2,4-D-amine	Glyphosphate
No surfactant	95	27	21
Allinol	97	40	9
Mon 0818	95	44	82
Oxysorbic	92	43	71
R-11	94	40	26

[Source: Harbour, *et al.*, *Weed Science*, (2003), **51**, page 430]

- (a) Suggest which herbicide is most effective in controlling Kochia. Explain your answer. [2]

- **Bromoxynil;**
- **with no surfactant;**
- **% biomass reduction is 95%;**
- **vs. 27% with 2,4-D-amine and 21% with glyphosphate;**
- **3.5X better than 2,4-D-amine and 4.5X better than glyphosphate;**

[Max 2]

(b) Using the data in table 3.1, discuss the advice that should be given to farmers on the use of surfactants. [4]

- **surfactants may help some herbicides in weed control;**
- **no effect of surfactant when using bromoxynil;**
- **surfactant helped with 2,4-D and glyphosphate;**
- **all surfactants increased % biomass reduction by 2 times with 2,4-D;**
- **glyphosphate was helped with by MON 0818 and oxysorbic;**
- **4 times increase with the use of MON 0818 and oxysorbic;**
- **R-11 has almost no effect;**
- **allinol interferes with/reduces effect of glyphosphate;**
- **9% vs 21% without surfactant;**

[Max 4]

Dinitroaniline herbicides are used for the selective control of weeds in broad-leaved crops. Long-term use of these herbicides has resulted in the appearance of resistant weeds.

(c) Explain how herbicide resistance may arise and spread in a weed population. [2]

- **Variation exists in the weed population;**
- **Over time, mutations of genes;**
- **leading to changes in offspring phenotypes such they are not susceptible to a normally lethal dose of herbicide;**
- **resistant weeds are at selective advantage;**
- **survive to reproduce resistant offspring;**
- **pass on favorable alleles;**
- **Eventually, with continued herbicide usage, the majority of the weed population will be resistant;**

[Max 2]

Dinitroaniline herbicides act by binding to the protein tubulin, thereby disrupting the formation of the spindle fibres and other cellular microtubules. Weeds resistant to these herbicides may carry a mutation in the gene of tubulin such that, at one site in the protein, the amino acid isoleucine replaces the normal amino acid threonine.

Another experiment was performed on maize cells to see whether the mutation in the tubulin gene found in weed was actually responsible for dinitroaniline resistance. Maize cells were grown in tissue culture and divided into four groups (M2, M1, N and C) which were treated as follows:

M2	: genetically engineered to express two copies of the mutated tubulin gene from weed
M1	: genetically engineered to express one copy of mutated tubulin gene from weed
N	: genetically engineered to express the normal tubulin gene from weed
C	: not genetically engineered

Each group of cells was then grown into callus.

(d) Describe briefly how plant cells may be grown into callus. [3]

- **surfaces of explants/shoot tip/ root tip containing meristematic (growing) cells;**
- **are sterilized using dilute sodium hypochlorite/bleach;**
- **and excised;**
- **and are transferred aseptically;**
- **to agar plate/ petri dishes/culture vessels;**
- **containing nutrients;**
- **and plant growth substances/plant growth regulators/cytokinin;**
- **to stimulate growth to form callus tissue by mitosis;**

[Max 3]

Each maize callus was then grown in the presence of different concentrations of dinitroaniline herbicide. The results are shown in Fig. 3.1 in which the shaded circles show the relative size of each piece of callus tissue after growth in the presence of herbicide.

maize callus	dinitroaniline herbicide concentration (mg dm^{-3})				
	0.00	0.05	0.10	0.50	1.00
M2					
M1					
N					
C					

Fig. 3.1

(e) Explain why maize callus C was used in this investigation. [1]

- As control;
- to compare if mutation in the tubulin gene found in weed was responsible for dinitroaniline resistance;;

(f) With reference to Fig 3.1, assess whether the mutated tubulin gene from weed is responsible for dinitroaniline resistance. Justify your answer. [3]

- mutated tubulin gene from weed is responsible for dinitroaniline resistance;
- callus size of M2 did not change;
- with increasing concentration of dinitroaniline / $0.00\text{--}1.00 \text{ mg dm}^{-3}$ dinitroaniline;
- Contains two copies of the mutated tubulin gene from weed;
- dinitroaniline herbicides is unable to bind to the protein tubulin, no disruption to the formation of the spindle fibres and other cellular microtubules / mitosis can continue;

[max 2]

- as compared to maize callus N/C;
- callus size shrank with increasing concentration of dinitroaniline / $0.00\text{--}1.00 \text{ mg dm}^{-3}$ dinitroaniline;
- no mutated tubulin gene from weed;
- dinitroaniline herbicides is able to bind to the protein tubulin, disruption to the formation of the spindle and other cellular microtubules / no mitosis;

[max 2]

OR

- as compared to maize callus M1 (where 1 mutated tubulin gene is absent);
- callus size shrank with increasing concentration of dinitroaniline / $0.00\text{--}1.00 \text{ mg dm}^{-3}$ dinitroaniline;

OR

- at dinitroaniline herbicide concentration of 1.00 mg dm^{-3} ;
- callus size of all the other maize callus – N, C and M1 shrank;

[Total: 15]

Write your answers to this question on the separate answer paper provided.
Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.
Your answers must be in continuous prose, where appropriate.
Your answers must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

- 4 (a) Discuss the advantages and disadvantages of micropropagation / plant tissue culture. [10]

Advantages [max 6]

1. Rapid multiplication;

- shoots produced during cloning produces buds which can be used to generate more shoots (using the same tissue culture technique);
- By recycling buds in this way, the number of plants produced is multiplied (at each stage);
- Plants with desired traits can be multiplied rapidly than can be done using conventional breeding methods/methods that rely on sexual reproduction;

[max 1]

2. Genetic uniformity;

- Since plants produced are genetically identical - possess the desirable features of the stock/parent plants;
- difficult to produce plants that breed true/homozygous for desired traits when sexual reproduction is used;

[max 1]

3. Production of disease-free plant material;

- viruses do not normally infect/penetrate to the tips of meristems/meristematic tissue;
- possible to use meristems/meristematic tissue for producing virus-free plants by tissue culture.
- plants are prepared in sterile conditions, they are free from surface bacteria and fungi that cause disease;

[max 1]

4. Take up little space;

- Compared with plants growing in fields, tissue culture takes up relatively little space;

5. Ability to air freight large quantities;

- Air freighting is cheaper because the cultures are not bulky;

6. Independent of climate changes;

- Tissue culture can be carried out independently of seasonal changes in climate;
- Plants can be produced out of season when they can be sold for higher prices;

[max 1]

7. Effective means of asexual reproduction in some plant species;

- Some plants like banana are sterile and have to be propagated asexually/Some plants like orchids have seeds that are difficult to germinate;
- These plants can be produced more reliably by asexual means;

[max 1]

8. Genetic engineering;

- Tissue culture can be used to produce transgenic plants with improved traits;

Disadvantages

9. Difficulty in maintaining sterile environment;

- Many plant species are easily infected by bacteria, even in very clean environments, contamination can cause very high losses in a short time;

10. High production costs;

- requires sophisticated facilities, sterile laboratory conditions and special nutrient media, trained personnel with specialized skills; (2 out of 4)

11. Labour intensive;

- A lot of labour is required to transfer the plantlets from the laboratory to the field;

12. High level of somatic variation;

- Callus cultures sometimes undergo genetic changes, only some of these changes are commercially useful;

13. Vulnerable to diseases;

- the clones are genetically identical, crops are very susceptible to new diseases/changes in environment;
- Whole crops could be wiped out;

[max 1]

- (b) Discuss the social and ethical implications of production of genetically modified organisms to solve the demand for food in the world. [10]

1) Definition of GMO

- **Genetically Modified Organism:** Organism that has acquired one or more genes by artificial means;
- The genes may or may not be from the same species;

2) Social implications

(a) Threat to Human Safety [max 3]

(i) Genetic markers;

- Vectors used for transforming plant/crop cells – contain antibiotic markers/kanamycin resistance;;
- example is Flavr Savr tomatoes;
- Concern that the antibiotic resistant gene (if it is not properly broken down by the digestive system) may be passed to potentially harmful bacteria making them resistant to the antibiotics;;

(ii) Allergens or Toxic/carcinogenic;;

- Transgenic food can be potential sources of allergens;;
- Eg. those in peanuts can cause allergic reactions in some people;;

(b) Threat to Safety of the Environment [max 3]

(i) Genetically modified crops might establish themselves as weeds;

- transgenic crops have been modified to withstand unfavourable environmental conditions (may be carried by the wind to other places);
- These crops grow quickly and may established themselves as weeds (award once);
- Spread of resistance from genetically modified crops to weeds;;
- Pollen grains from genetically modified plants can be carried in the wind and fertilise wild relatives and weeds;;
- eg. Herbicide resistant plant;
- result in “superweeds” – weeds that are resistant to herbicide;
- become more invasive (overrun agricultural areas rapidly) / compete with the natural species;

(ii) Upsetting the ecological balance;;

- genetically modified organisms themselves may be more aggressive thus they may upset the balance of the native species;;
- eg. Giant salmon;
- decrease in biodiversity;;

(iii) Use of pesticide;;

- developing pesticide-resistant plants may encourage the use of greater amounts of pesticides;;
- possibility of pests developing resistance to toxin produced by crop plants;;
- eg. Bt toxin;

3) Ethical implications [max 3]

- (a) There is concern about infringing nature when genes of plants and animals are tampered with by mixing genes among species;;
- (b) The issue of engineering any organism to be used as food for human consumption might not be acceptable by all;;
 - Eg, the use of BST in dairy cattle increases the risk of mastitis. There seems to be *little concern about whether the animals are biologically 'designed'* to withstand the additional burden of increased milk or egg or meat production.
- (c) Religious concerns or dietary restrictions;
 - Eg. Plants that have been modified to carry animal genes or vice versa;
- (d) There is concern about the rights of patenting a genetically modified animal or plant;;
- (e) Companies have sought to patent the transgenic plants or animals that they have developed;
Some people argued that patenting animals is itself unethical as it reduces them to the level of objects;
If patenting is not allowed, a company might not be able to protect the results of their research program.
- (f) Using genetic engineering to make a staple crop more resist in marginal conditions (e.g. drought, cold) could be a potential boom for Third World agriculture but could also possibly be another danger of increased dependence on rich "developed" countries.
 - i.e. richer countries benefiting at the expense of the poorer countries;
 - world food production may be dominated by a small number of large companies with the technical knowledge;
- (g) The potential risks and possible consequences involved in releasing genetically modified organisms into the biosphere are not known;;
- (h) Labeling of products on sale to indicate that genetic engineering was involved in their production is not mandatory in many countries;;